

Relation-Aware Multimodal Analogical Reasoning with Modality Fingerprints and Adaptive Gating

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Abstract

Analogical reasoning over Multimodal Knowledge Graphs (MMKGs) couples abductive relation induction with inductive tail completion. However, existing approaches rely on static fusion mechanisms that overlook the inherent asymmetry of modal relevance: while visual cues elucidate concrete entities, they are often noisy or irrelevant for abstract concepts, where text and graph structure provide decisive signals. Furthermore, prior methods fail to enforce consistency between induced relations and the modality patterns implied by the analogical context. To bridge this gap, we introduce **RMAR**, a Relation-aware Multimodal Analogical Reasoning framework with two complementary paths. An explicit path estimates *modality fingerprints* to score compatibility during relation induction and guide fusion during tail completion. An implicit path employs adaptive gating to blend structural, textual, and visual signals conditioned on the specific query context. To address the limitations of current benchmarks, which overrepresent concrete entities, we release **MCNetAnalogy**, and its companion graph, **MCNetKG**, a rigorous dataset enriched with abstract concepts and actions. RMAR is backbone-agnostic and works with multimodal knowledge graph embedding (MKGE) and transformer-based (MPT) pipelines. Extensive experiments demonstrate that RMAR delivers consistent gains across both embedding-based and transformer-based backbones, achieving a 29% relative improvement on MCNetAnalogy. Ablation studies confirm that RMAR's relation-aware modulation is particularly effective when modal evidence is weak or ambiguous.

CCS Concepts

• Computing methodologies → Knowledge representation and reasoning.

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Keywords

Analogical Reasoning; Multimodal Knowledge Graph; Relation Prediction; Link Prediction; Knowledge Representation Learning

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Resource Availability:

The source code and data of this paper has been made publicly available at <https://doi.org/10.5281/zenodo.18346949>.

1 Introduction

Analogy is a core mechanism of human cognition [6, 11, 16, 19, 38]. For instance, a singer performs a song as a violinist performs a concerto; the performative relation aligns while surface cues differ. The analogical mechanism enables people to solve novel problems by mapping structured relations from a familiar domain to a new one, thus supporting transfer, explanation, and decision making [11, 13, 16, 17, 19]. Classic theories and empirical results across structure mapping, mental models, and cognitive studies converge on the view that analogy depends on aligning relational structure rather than matching surface features [11]. Meanwhile, knowledge graphs provide an explicit relational substrate that aligns with this view and already underpins recommendation, question-answering, and knowledge discovery [3, 18, 34]. Incorporating multiple modalities, such as text and images, yields a more comprehensive and human-like representation of entities and relations [9], enabling analogical reasoning that aligns with relational structure while drawing on complementary semantic and perceptual evidence.

Analogical reasoning over multimodal knowledge graphs (MMGs) couples *abductive* relation induction with *inductive* tail completion. To illustrate, consider (*arsenic*, *very_dangerous*) : (*cyanide*, ?) in Figure 1. The correct tail is “*very_dangerous*”, yet the visual modality is weak or misleading because the images of the crystals or vials do not convey hazard. In contrast, textual definitions and graph structure (e.g., *is_a*(*chemical_compound*), *causes*(*poisoning*)) provide

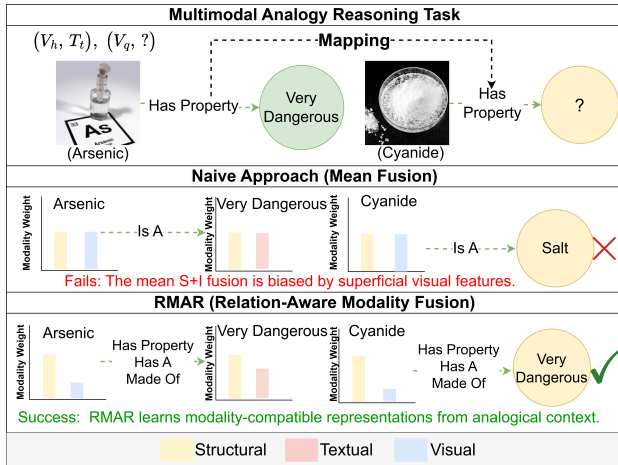


Figure 1: Top: analogical transfer from (Arsenic, Very Dangerous) to (Cyanide, ?). Middle: mean fusion is skewed by visual bias and fails. Bottom: RMAR accounts for modality compatibility and recovers the correct tail Very Dangerous.

discriminatory signals. This mirrors human use of abstract parallels in a multimodal setting where relations must be inferred from both structured and perceptual evidence. The core challenge in the analogical reasoning of MMKG is to infer the latent relation “*has_property*” from (*arsenic*, *very_dangerous*), the *analogical context*, without supervision, and then to determine the contribution of each modality to recover the missing entity in the *query context*.

Despite progress in multimodal knowledge graph embedding (MKGE) and transformer-based pipelines (MPT) [3, 9, 36, 38], current studies face three limitations that are critical for analogical reasoning. First, fusion is often fixed or shallow, without modeling relation-dependent preferences or head-and-tail modality preferences asymmetry; thus, these models cannot adapt when a relation is better expressed by graph topology than by image or text [9, 36, 38], like Arsenic or Cyanide in Figure 1. Second, existing abductive relation induction methods typically identify a *single plausible relation*, using either a plausibility score or a shared relation representation, without an explicit test that the inferred relation matches the modality pattern of the analogical context. This introduces ambiguity and unstable selection when multiple relations are plausible [3, 29, 30]. Third, the benchmark coverage is skewed: Wikidata-derived datasets overrepresent concrete, nameable entities (e.g., *wine*, *egg*) and underrepresent abstract concepts and actions. As a result, existing benchmarks primarily test visual and textual recognition on concrete objects but do not adequately assess whether methods can reason about abstract properties (e.g., “*very_dangerous*”), where visual evidence is intrinsically weak. This imbalance limits the rigor of evaluation [28, 38]. These gaps degrade performance whenever one modality is weak or ambiguous, leaving open the central question of how to align abductive relation prediction with inductive tail completion under relation-aware and modality-aware evidence.

To overcome these challenges, we present **RMAR**, a **R**elation-aware **M**ultimodal **A**nalogical **R**easoning framework. RMAR learns

modality-compatible representations from analogical context, both explicitly through relation-level head-and-tail fingerprints and implicitly through gated fusion, and then inductively completes the missing tail inference. RMAR consists of two coupled reasoning phases: (i) the *abductive reasoning phase* (i.e., relation induction) and (ii) the *inductive reasoning phase* (i.e., tail completion). In the abductive phase, RMAR explicitly estimates head-and-tail modality fingerprints for each relation and scores each candidate based on modality compatibility, which measures the agreement between the modality pattern from the analogical context and each relation fingerprint profile. Then RMAR retains the top-K relations. Implicitly, an adaptive gate in the encoder, conditioned on the query and the analogical context, blends structural, textual, and visual features to form context-aware embeddings that preserve relation-aware evidence. In the inductive phase, the explicit path performs a relation-guided fusion between the query head and each candidate tail for the retained top-K relations using the corresponding fingerprints. It aggregates evidence across relations with soft weights to produce final tail scores. The implicit alternative directly utilizes the gated encoder representations, which preserve salient evidence from the query and the analogical context to infer the missing tail.

Note that RMAR is agnostic to MKGE and MPT backbones. With MKGE, one can fine-tune embeddings to align with relation fingerprints, allowing explicit relation induction and tail completion to reinforce each other. With MPT, one can fine-tune the encoder to align its gated representations, conditioned on the query and the analogical context, improving both abductive and inductive inference. To address data limitations, we release MCNetANALOGY and its companion multimodal knowledge graph MCNetKG. Derived from ConceptNet analogies, MCNetANALOGY extends beyond concrete entities to include a rich set of abstract concepts and actions, broadening coverage and increasing task difficulty for multimodal analogical reasoning. Our contributions are summarized as follows:

- We present RMAR, a principled paradigm for multimodal analogical reasoning that unifies an explicit fingerprint-based path and an implicit gated fusion alternative across dominant backbones, providing a solid foundation for future research in this setting through the new benchmark and knowledge graph.
- *Explicit fingerprint-based Approach.* For each relation, RMAR estimates head and tail modality fingerprints and uses them to (i) score modality compatibility in abductive relation induction and (ii) guide fusion in inductive tail completion, enforcing consistency with the analogical context.
- *Implicit gating-based Approach.* Learn a relation-aware gated fusion inside the encoder, conditioned on the query and the analogical context, that blends structure, text, and vision into context-aware embeddings, preserving salient evidence and reducing modality blind spots to strengthen both phases.
- We release MCNetANALOGY and companion multimodal knowledge graph MCNetKG, which adds a rich set of abstract concepts and actions and addresses a critical gap in prior datasets. Extensive experiments on MCNetANALOGY and MARS with MKGE and MPT backbones show consistent reasoning gains, including a relative improvement of up to 29% on MCNetANALOGY.

2 Related Works

Unimodal Analogical Reasoning. Analogical reasoning has been studied extensively in both vision and language, yet typically within a single modality. In computer vision, benchmarks such as PGM [1] and RAVEN [37] task models with completing visual patterns, sometimes augmented by graph-based object relations [14, 15], but ignore complementary textual semantics. In natural language, word-analogy tests (“king:queen : man:woman”) leverage vector arithmetic [23, 24] or contextual embeddings from large pre-trained models [4, 10] to reveal semantic regularities, yet they neither chain multistep relations nor incorporate visual or structured cues. As a result, vision-only methods excel at geometric or attribute transformations but lack linguistic grounding, while text-only approaches capture rich meanings but cannot exploit visual patterns or explicit relational structure. This gap motivates our work in integrating visual, textual, and graph-structured information into a unified, multimodal framework for analogical reasoning.

Multimodal Knowledge Graph Embedding (MKGE). Although MKGE methods do not directly tackle analogical reasoning, they can be adapted by decomposing the problem into **relation prediction** (abduction) followed by **tail prediction** (induction). Early MKGE work focused on static fusion of visual or textual features into classic KG embeddings. IKRL [36] learns CNN-derived image representations for each entity and incorporates them through an attention-weighted sum into a TransE scoring function. TransAE [35] pre-trains a joint auto-encoder over text and image features to project both modalities into a unified space, then applies TransE for scoring. RSME [33] introduces a relevance-based module that filters out noisy or uninformative images by assigning modality-specific confidence scores before embedding fusion. More recently, MoSE [39] has employed self-attention to model fine-grained interactions between text, vision, and structure, dynamically weighting each modality when computing triple scores. While these approaches demonstrate clear benefits over structure-only baselines, they generally rely on fixed or shallow fusion schemes and do not fully capture heterogeneous modality preferences at the level of individual entities or relations, a gap that our model aims to address.

Multimodal Pretrained Transformer (MPT). Multimodal pretrained transformers (MPTs) can be broadly classified as single-stream (e.g., VisualBERT [21], ViLT [20]), which process interleaved text and image tokens in one transformer; dual-stream (e.g., ViLBERT [22]), with separate vision and language encoders connected via cross-modal attention; and mixed-stream (e.g., FLAVA [27], MKGFormer [7]), which combines shared layers with modality-specific components. Recent work [5] adds triplet-level modality interactions for stronger inference. However, like prior MPT and MKGE methods, it still relies on a single, general relation embedding. It cannot leverage a high-confidence set of candidate relations to boost prediction accuracy further.

3 Problem Formulation

Definition 3.1 (Multimodal Knowledge Graph). A multimodal knowledge graph (MMKG) is a tuple $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T}, \mathcal{M})$, where $\mathcal{E}$ is the entity set, $\mathcal{R}$ is the relation set, $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$ is the set of observed triples, and $\mathcal{M} = \{S, V, L\}$ is the modality set comprising structural (S), visual (V), and Textual (T) views.

Definition 3.2 (Analogical Query). An analogical query on an MMKG $\mathcal{G}$ is the colon-form pair $Q = (e_h, e_t) : (e_q, ?)$, where $e_h, e_t, e_q \in \mathcal{E}$. The ordered pair (e_h, e_t) is the *analogical context*, and $(e_q, ?)$ is the *query context* with e_q the *query head*.

Problem (Analogical Reasoning over MMKG). Given an MMKG $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T}, \mathcal{M})$ and an analogical query $Q = (e_h, e_t) : (e_q, ?)$, the *analogical reasoning over $\mathcal{G}$* is to find a tail $e_a \in \mathcal{E}$ such that the latent relation expressed by (e_h, e_t) is applied analogously to (e_q, e_a) .

Crucially, this latent relation is not provided and must be inferred from the features of the entities. Based on the availability of modalities, we distinguish two reasoning settings.

- **Single Analogical Reasoning:** The pair of entities in the analogical context is homogeneous in modality. That is, both e_h and e_t are in the visual domain or both in the textual domain, while the query head e_q may be one of the modalities. This setting challenges intra-modal relational transfer, such as $(V_h, V_t) : (V_q, ?)$ or $(T_h, T_t) : (T_q, ?)$, and cross-modal relational transfer, such as $(V_h, V_t) : (T_q, ?)$ or $(T_h, T_t) : (V_q, ?)$.
- **Blended Analogical Reasoning:** The pair of entities in the analogical context is heterogeneous in modality, and the query head e_q may be one of the modalities. This setting reflects a real-world setting, such as $(V_h, T_t) : (V_q, ?)$, which requires the model to infer relations from a heterogeneous modality.

4 Methodology

4.1 Framework Overview

We introduce **RMAR**, a **Relation-aware Multimodal Analogical Reasoning** framework. Given an analogical query $Q = (e_h, e_t) : (e_q, ?)$ over an MMKG $\mathcal{G}$, RMAR consists of two complementary approaches that infer the missing tail. Both share the same initialization (§4.2). Figure 2 illustrates the architecture of RMAR.

- (1) *Explicit Fingerprint-based Approach* (§4.3). The abductive phase performs explicit relation induction guided by modality fingerprints that capture entity-level preferences and relation-level head–tail asymmetry, resulting in a ranked set of relation candidates. The inductive phase leverages these fingerprints to guide tail completion across the selected candidates.
- (2) *Implicit Gating-based Approach* (§4.4). The abductive phase learns adaptive gated fusion within the transformer encoder to integrate evidence from both the analogical and query contexts, resulting in context-aware representations. The inductive phase leverages these representations to infer the most plausible tail by analogical transfer.

4.2 Multimodal Embedding Initialization

To leverage the complementary information in each modality, we initialize entity representations with encoders that transform raw features into modality-specific embeddings, a standard practice in multimodal knowledge graph settings. For an entity $e \in \mathcal{E}$ and a modality $m \in \mathcal{M}$, the initialization is given by:

$$\mathbf{h}_e^{(m)} = \text{PE}_m(x_e^{(m)}), \quad (1)$$

where $x_e^{(m)}$ denotes the raw input of e in modality m (for example, a local graph neighborhood, an image, or a text description), and

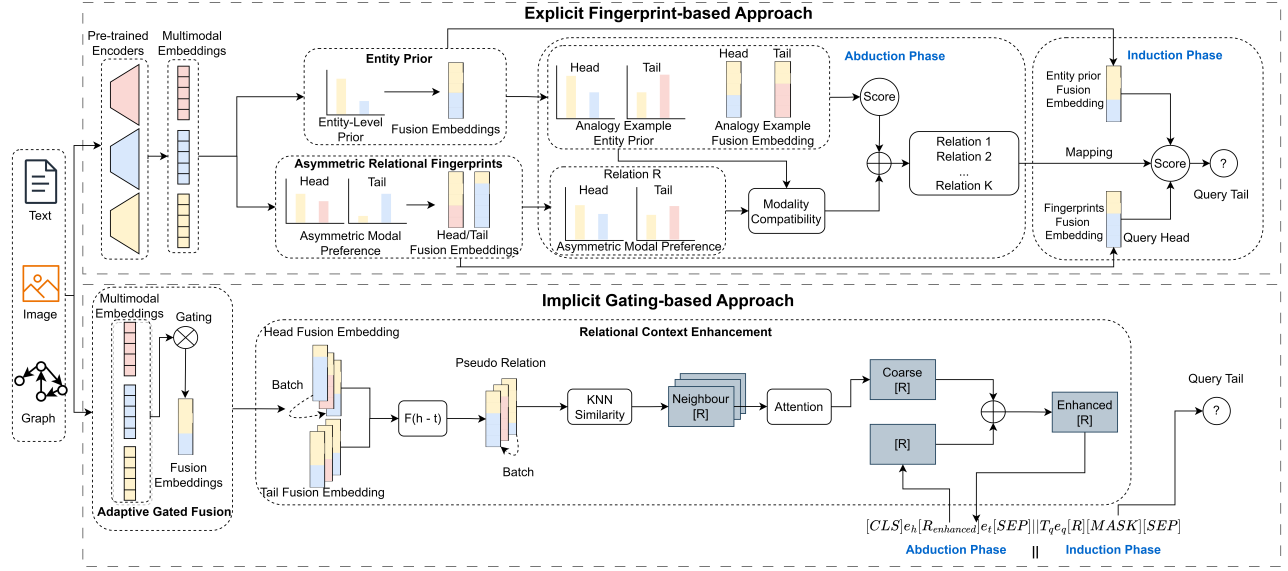


Figure 2: Architecture of the Relation-Aware Multimodal Analogical Reasoning (RMAR) framework.

PE_m is the corresponding pre-trained encoder to extract modality-specific semantic features.

The specific encoders are selected according to the backbone family as follows. For the MKGE family, we utilize VGG16 [26] for visual features, BERT [10] for textual features, and Graph Attention Network (GAT) [32] for structural features. For the MPT family, we utilize the pre-trained CLIP image encoder [25] for visual features, BERT for textual features, and Relphormer [2] for structural features. By default, the visual and textual encoders PE_{visual} and PE_{text} act as frozen feature extractors, preserving broad semantic knowledge learned during pre-training, while structural embeddings are initialized as learnable parameters to capture graph-specific topology. In the MPT setting, we initialize from the same encoders and fine-tune them within our objectives when indicated, so that gated representations align with the analogical context. This results in stable semantic features of vision and text, allowing the structure to adapt to the target graph and providing a coherent initialization for both explicit and implicit approaches of RMAR.

4.3 Explicit Fingerprint-based Approach

4.3.1 Abductive Relation Induction Phase. Not all modalities contribute equally across entities or relations. For abstract properties such as toxicity, visual evidence is weak while text and structure carry decisive cues; for fine-grained categories, vision can exceed text. Many relations are asymmetric in role, with heads and tails requiring different evidence. The explicit variant of RMAR therefore aims to learn (i) an **entity prior** over modalities that captures persistent, type-dependent signal strength, and (ii) **asymmetric relational fingerprints** that encode asymmetry between head-and-tail modalities. In inference, the prior is context-conditioned by the analogical query, allowing the same entity to shift emphasis given the example pair.

Entity Prior. We model an entity-level prior as a weight vector that reflects its persistent preference over modalities. Formally,

given modality specific embeddings $\{\mathbf{h}_e^{(m)} \mid m \in \mathcal{M}\}$ for entity e , we adapt the Dynamic Cross modal Weighted (DCMW) [8] to estimate normalized modal preference $\{w_e^{(m)}\}$ for all $m \in \mathcal{M}$:

$$w_e^m = \text{DCMW}(\{\mathbf{h}_e^{(m)}\}), \quad m \in \mathcal{M}. \quad (2)$$

Once the prior weights are obtained, the representation of the fused entity $\mathbf{h}_e$ is a preference-weighted sum as follows:

$$\mathbf{h}_e = \sum_{m \in \mathcal{M}} w_e^m \cdot \mathbf{h}_e^{(m)}, \quad (3)$$

which allows the model to adaptively emphasize structure for abstract concepts and vision or text for concrete entities.

Asymmetric Relational Fingerprints. To encode asymmetric modal preferences between heads and tails, each relation maintains role-specific fingerprints that specify the expected evidence for heads and tails under $r \in \mathcal{R}$. In the abductive relation induction phase, RMAR favors candidates whose active modalities align these fingerprints, reducing ambiguity when the modalities disagree. Beyond its primary semantic embedding $\mathbf{h}_r$, each relation exhibits modal preferences across roles; we capture this asymmetry by learning two preference priors per relation, $\mathbf{w}_{rh}$ and $\mathbf{w}_{rt}$.

Formally, for each relation $r \in \mathcal{R}$, RMAR estimates two distributions on the modalities, one for the heads $\mathbf{w}_{rh} \in \mathbb{R}^{|\mathcal{M}|}$ and one for tails $\mathbf{w}_{rt} \in \mathbb{R}^{|\mathcal{M}|}$, where $|\mathcal{M}|$ is the number of modalities. The ideal head prior is the average of the entity preferences in the dataset.

$$\mathbf{w}_{rh} = \frac{1}{|H_r|} \sum_{e_h \in H_r} \mathbf{w}_{e_h}, \quad (4)$$

where H_r is the set of head entities observed with r . Direct computation over the full graph is computationally expensive, so we maintain online estimates using an Exponential Moving Average

(EMA). For a batch of triples B , we compute a batch-local estimate:

$$\hat{\mathbf{w}}_{rh}^{(B)} = \frac{1}{|H_r^{(B)}|} \sum_{e_h \in H_r^{(B)}} \mathbf{w}_{e_h}, \quad (5)$$

where $H_r^{(B)}$ is the set of head entities for the relation r within batch B . We then update the global prior $\mathbf{w}_{rh}$ as follows:

$$\mathbf{w}_{rh} \leftarrow \beta \cdot \mathbf{w}_{rh} + (1 - \beta) \cdot \hat{\mathbf{w}}_{rh}^{(B)}, \quad (6)$$

where $\beta \in [0, 1)$ is a momentum hyperparameter. A similar update is performed for the tail entity to obtain global prior $\mathbf{w}_{rt}$.

Given the induced relation $\hat{r}$ from the explicit abductive phase, RMAR constructs a relation-aware embedding for the query head q_h by using the head prior $\mathbf{w}_{\hat{r}h}$ as fusion weights:

$$\mathbf{h}'_{q_h} = \sum_{m \in \mathcal{M}} \mathbf{w}_{\hat{r}h}^{(m)} \cdot \mathbf{h}_{q_h}^{(m)}. \quad (7)$$

An analogous fusion, using the tail prior $\mathbf{w}_{\hat{r}t}$, is applied to each candidate tail entity:

$$\mathbf{h}'_{e'} = \sum_{m \in \mathcal{M}} \mathbf{w}_{\hat{r}t}^{(m)} \cdot \mathbf{h}_{e'}^{(m)}. \quad (8)$$

These asymmetric relational fingerprints determine how the modalities are combined, ensuring that the resulting representations are tailored to the specific relation induced.

Modality Compatibility. For the MKGE family backbones, the model must explicitly induce a latent relation r from an example pair (e_h, e_t) in an analogical context, typically by maximizing a structural score (e.g., RotatE or DistMult). However, since there is no direct supervision of r , induction can drift toward relations that are structurally plausible but modally inconsistent with the evidence in the example. We address this with a modality compatibility term that rewards relations whose asymmetric relational fingerprints align with the observed modal preferences of the example.

Formally, given the entity priors $\{\mathbf{w}_e^m \mid m \in \mathcal{M}\}$ and the asymmetric relational fingerprints $(\mathbf{w}_{rh}, \mathbf{w}_{rt})$ for an analogical example pair (e_h, e_t) , we calculate a *modality likelihood* $s_{\text{modality}}(r)$ for each candidate relation $r \in \mathcal{R}$. This score evaluates the probability that a candidate relation r will generate the modal preferences observed in the example pair (e_h, e_t) . Specifically, we treat the learned relational priors $\mathbf{w}_{rh}$ and $\mathbf{w}_{rt}$ as parameters of two preference distributions. Then, the likelihood of observing the entity preference vector $\mathbf{w}_{e_h}$ is given by the probability function:

$$p(\mathbf{w}_{e_h} | \mathbf{w}_{rh}) = \prod_{m \in \mathcal{M}} (\mathbf{w}_{rh}^{(m)})^{\mathbf{w}_{e_h}^{(m)}}. \quad (9)$$

This structure ensures that modalities with higher weight in the entity's profile contribute more to the final likelihood score, strongly rewarding alignment between the entity's observed pattern and the relation's expected pattern.

The total modality likelihood for the pair is the product of the head and tail likelihoods:

$$s_{\text{modal}}(r) = p(\mathbf{w}_{e_h} | \mathbf{w}_{rh}) \times p(\mathbf{w}_{e_t} | \mathbf{w}_{rt}) \quad (10)$$

We combine modality compatibility with the structure-based plausibility $s_{\text{structure}}(e_h, r, e_t)$ from the MKGE scorer (e.g., RotatE or DistMult) as follows:

$$s_{\text{final}}(r) = s_{\text{structure}}(e_h, r, e_t) + \log(s_{\text{modal}}(r)). \quad (11)$$

We use the log-likelihood for numerical stability and to convert the multiplicative probabilities into an additive form that combines more naturally with distance-based structural scores. As a result, a relation $\hat{r}$ is induced by maximizing the modality likelihood:

$$\hat{r} = \arg \max_{r \in \mathcal{R}} s_{\text{final}}(r). \quad (12)$$

Account for both the likelihood of the modality and the structural plausibility; the explicit variant of RMAR prefers relations that are structurally plausible and modally consistent with the evidence profile of (e_h, e_t) , capturing the asymmetric relational fingerprints of the head and tail while remaining fully differentiable.

4.3.2 Inductive Tail Completion Phase. Selecting a single relation can be unstable when several candidates receive similar scores. We therefore aggregate evidence from the top K relations ranked by the joint score $s_{\text{final}}(r)$.

Relation Confidence. Let the set of top- K candidate relations be $\mathcal{R}_K$, formed by inducing the relations with the highest scores $s_{\text{final}}(r)$ with the hyperparameter K . For each $r' \in \mathcal{R}_K$, the confidence for $\omega_{r'}$ is calculated by applying a Softmax function to the original prediction scores of the top- K relations. This gives more influence to the relations of higher confidence as follows:

$$\omega_{r'} = \frac{\exp(s_{\text{final}}(r'))}{\sum_{r^* \in \mathcal{R}_K} \exp(s_{\text{final}}(r^*))} \quad \forall r' \in \mathcal{R}_K. \quad (13)$$

Multi-Relation Aggregation. For each candidate tail $e' \in \mathcal{E}$, the final score for e' , denoted as $s_{q_t}(e')$, is the weighted sum of the individual structural scores computed from the MKGE scorer (e.g., RotatE or DistMult) as follows:

$$s_{q_t}(e') = \sum_{r' \in \mathcal{R}_K} \omega_{r'} \cdot s_{\text{structure}}(q_h, r', e'), \quad (14)$$

where $s_{\text{structure}}$ is the scoring function of the MKGE model.

By combining relation confidence with soft aggregation over the top- K candidates, the explicit variant of RMAR is less susceptible to noise in relation induction and yields a more robust contextually grounded basis for tail completion.

4.4 Implicit Gating-based Approach

Given an analogical context (e_h, e_t) and a query context $(e_q, ?)$, an analogical MPT asks to recover the missing entity such that the transformation expressed in the analogical context transfers to the query. To this end, the objective of the implicit variant of RMAR is to learn (i) **context-aware representations** that implicitly preserve the analogical context while remaining sensitive to the query, and (ii) **context enhancement** that stabilizes these representations, thus improving generalization in inductive tail completion.

4.4.1 Adaptive Gated Fusion. An analogical MPT query transfers a *relation effect* from an analogical context to a query context. The fused representation must be *query-aware*, which attends to the most reliable modality for the query context; and *analogy preserving*, which shapes the representation so that the relation effect encoded by the analogical context remains usable as a template for the query head by landing near the correct tail.

To this end, we introduce an *Adaptive Gated Fusion* to dynamically learn the preference for each modality within a specific context. Specifically, we adopt Relphormer [2] to obtain a structural

Dataset	Train/Dev/Test	Entity	Relation	Triple	Images	Base	Characteristic
MCNetAnalogy	1,660/208/207	1,663	18	2,304	8,115	Conceptnet Relational Similarity [31]	Abstract, Action, Concrete Entities
MARS [38]	10,641/1,362/1,020	2,063	27	13,023	13,398	EKAR [6], BATS [12]	Concrete Entities

Table 1: Statistical information of the Multimodal Analogy Knowledge Graphs in our benchmark.

embedding $\mathbf{h}_e^{(s)}$ for each entity e via link prediction pre-training on MCNetKG and MarKG. To determine how to balance multi-modal signals, using the fusion of structural embedding $\mathbf{h}_e^{(s)}$ and textual embedding $\mathbf{h}_e^{(t)}$ as an example, the gating mechanism first computes a dynamic gate vector:

$$\mathbf{g}_e = \sigma(\mathbf{W}_g[\mathbf{h}_e^{(t)}; \mathbf{h}_e^{(s)}] + \mathbf{b}_g), \quad (15)$$

where $[\cdot; \cdot]$ denotes concatenation, $\mathbf{W}_g$ and $\mathbf{b}_g$ are learnable parameters, and σ is the sigmoid function. Meanwhile, a candidate transformation vector $\mathbf{h}^{(\text{trans})}$ is computed by passing both embeddings through a separate non-linear transformation:

$$\mathbf{h}^{(\text{trans})} = \tanh(\mathbf{W}_t[\mathbf{h}_e^{(t)}; \mathbf{h}_e^{(s)}] + \mathbf{b}_t), \quad (16)$$

where $\mathbf{W}_t$ and $\mathbf{b}_t$ are learnable parameters.

This adaptive gate then blends the original semantic signal with its context-conditioned transformation. The initial fused representation $\tilde{\mathbf{h}}_e$ is computed as:

$$\tilde{\mathbf{h}}_e = (1 - \mathbf{g}_e) \circ \mathbf{h}_e^{(t)} + \mathbf{g}_e \circ \mathbf{h}^{(\text{trans})}, \quad (17)$$

where $\circ$ represents the Hadamard (element-wise) product. Finally, for stabilization, we employ a residual connection to the original semantic embedding followed by Layer Normalization to produce the final fused output, $\mathbf{h}_e^{(\text{fused})}$.

Crucially, the gate vector $\mathbf{g}_e$ is not a static global parameter, as shown in Eq.(15), it is recomputed for each entity per query, so the model can upweight the stable structural prior when text is ambiguous and amplify text when definitions are discriminative. The exact mechanism fuses structure with vision by replacing $\mathbf{h}_e^{(t)}$ with visual information $\mathbf{h}_e^{(v)}$.

4.4.2 Relational Context Enhancement. Prior MPT-based approaches typically enforce relational consistency in a single instance [38], which yields a *fine-grained* alignment that is highly dependent on the specific context of a single instance, making the representation sensitive to linguistic variations. To overcome this, we instead pool evidence from multiple semantically similar relation instances to form a *coarse-grained*, more stable context-aware representation that is less sensitive to linguistic variation.

Pseudo Relation and Neighbor Retrieval. To measure the similarity between different relation instances, we generate a query-conditioned relational embedding, $\mathbf{r}_{\text{emb}}^{(i)}$, inspired by TransE, as the *pseudo relation* between the final hidden states of the head and tail entities ($\mathbf{r}_{\text{emb}}^{(i)} = \mathbf{h}_h^{(i)} - \mathbf{h}_t^{(i)}$). We use this to identify a set of top- K nearest neighbors $\mathcal{N}_i$ for each anchor instance i within the batch. **Neighborhood Attention.** RMAR then fuses the information from these neighbors with attention. For an anchor i , its fine-grained “[R]” context representation, $\mathbf{h}_{[\text{R}]}^{(i)}$, serves as the *Query*, while the representations of its neighbors, $\{\mathbf{h}_{[\text{R}]}^{(j)}\}_{j \in \mathcal{N}_i}$, serve as both *Keys*

and *Values*. This produces a coarse-grained context representation, $\mathbf{h}_{[\text{R}], \text{coarse}}^{(i)}$, as the weighted sum of the neighbors’ representations:

$$\mathbf{h}_{[\text{R}], \text{coarse}}^{(i)} = \text{Attention}(\mathbf{h}_{[\text{R}]}^{(i)}, \{\mathbf{h}_{[\text{R}]}^{(j)}\}_{j \in \mathcal{N}_i}). \quad (18)$$

Finally, we obtain the enhanced context representation via a residual connection, which integrates the aggregated neighborhood context with the original fine-grained representation:

$$\mathbf{h}_{[\text{R}], \text{enhanced}}^{(i)} = \mathbf{h}_{[\text{R}]}^{(i)} + \mathbf{h}_{[\text{R}], \text{coarse}}^{(i)}. \quad (19)$$

This enhanced context-aware representation, $\mathbf{h}_{[\text{R}], \text{enhanced}}$, encodes a coarse-grained notion of the relation, improving robustness for the subsequent scoring task and ultimately improving the final prediction. The procedure is efficient per instance after neighbor selection, which extends trivially to multi-head attention or a cross-batch memory to enlarge $\mathcal{N}_i$ when desired. See Appendix A.3 for an alternative neighborhood expansion strategy and its performance.

4.5 Training and Inference

4.5.1 Explicit Fingerprint-based Approach. The training objective for the MKGE family integrates signals from our dual fusion mechanism. It combines a standard MKGC loss calculated on entity-level fusion embeddings ($\mathcal{L}_{\text{entity}}$) with a loss calculated on the relation-level fusion embeddings ($\mathcal{L}_{\text{relation}}$). The final objective is their weighted combination:

$$\mathcal{L}_{\text{MKGE}} = \lambda \cdot \mathcal{L}_{\text{entity}} + (1 - \lambda) \cdot \mathcal{L}_{\text{relation}}, \quad (20)$$

where λ is a balance hyperparameter, and the inference procedure mirrors this objective. For each candidate entity, a final score is computed by taking the same weighted sum of the scores from the entity-level and relation-level views. The top-scoring entity is then selected as the final prediction. Algorithm 1 summarizes the explicit fingerprint-based approach.

4.5.2 Implicit Gating-based Approach. The MPT family is trained with a multi-task objective. The primary term is a cross-entropy loss $\mathcal{L}_{\text{mask}}$ for the masked entity prediction task, supplemented by relational consistency $\mathcal{L}_{\text{relation}}$ that aligns the enhanced relation token of the example with the relation token of the query:

$$\mathcal{L}_{\text{relation}} = 1 - \cos(\mathbf{h}_{[\text{R}], \text{enhanced}}^{(\text{ex})}, \mathbf{h}_{[\text{R}]}^{(\text{q})}). \quad (21)$$

The final training objective is a weighted sum of the two loss terms:

$$\mathcal{L}_{\text{MPT}} = \alpha \cdot \mathcal{L}_{\text{mask}} + (1 - \alpha) \cdot \mathcal{L}_{\text{relation}}, \quad (22)$$

where λ is a balancing hyperparameter. During inference, prediction is performed directly by selecting the entity with the highest logit at the “[MASK]” position. The benefit from our relational aggregation mechanism is implicit, as the training process enhanced the encoder with a more robust understanding of relational semantics, which in turn improves the accuracy of the final prediction. Algorithm 2 summarizes the implicit gating-based approach.

Table 2: Performance Comparison on MCNetAnalogy and MARS. Intra-family comparisons (MKGE vs. MKGE; MPT vs. MPT). Best per column is bold; second best is underlined. “–” indicates an unavailable result due to missing or incompatible code. The Relative Improvement row shows the percentage gain of our RMAR model over the second-best baseline. The relative improvement is calculated through RMAR and the suboptimal results.

Method Baselines		MCNetAnalogy					MARS				
		MRR	Hits@1	Hits@3	Hits@5	Hits@10	MRR	Hits@1	Hits@3	Hits@5	Hits@10
MKGE	TransAE	0.259	0.222	0.266	0.290	0.333	0.276	0.261	0.285	0.289	0.293
	IKRL	0.309	0.271	0.333	0.348	0.362	0.283	<u>0.266</u>	0.294	0.301	0.310
	RSME	<u>0.361</u>	0.295	0.396	0.435	0.502	<u>0.285</u>	<u>0.266</u>	0.298	0.307	0.311
	RMAR-RotatE	0.475	0.387	0.527	0.585	0.623	0.300	0.268	0.314	0.331	0.356
	Relative Improvement	+31.6%	+31.2%	+33.1%	+34.5%	+24.1%	+5.3%	+0.8%	+5.4%	+7.8%	+14.5%
MPT	VILT	0.240	0.165	0.261	0.306	0.404	0.266	0.245	0.275	0.287	0.303
	ViLBERT	0.340	0.239	0.386	0.465	0.537	0.292	0.256	0.312	0.327	0.347
	VisualBERT	0.435	<u>0.365</u>	0.476	0.503	0.547	0.284	0.261	0.292	0.308	0.321
	FLAVA	<u>0.441</u>	0.329	<u>0.500</u>	<u>0.567</u>	<u>0.656</u>	0.288	0.264	0.303	0.309	0.319
	MKGFormer	0.399	0.317	0.439	0.492	0.554	0.341	0.301	0.367	0.380	0.408
	TIMAR	-	-	-	-	-	<u>0.344</u>	<u>0.305</u>	-	0.383	0.416
	RMAR-MKGFormer	0.514	0.426	0.575	0.620	0.670	0.348	0.319	0.363	0.380	0.393
	Relative Improvement	+16.6%	+16.7%	+15.0%	+9.3%	+2.1%	+1.2%	+4.6%	-1.1%	-0.8%	-5.5%

Table 3: Ablation study of RMAR on the MCNet dataset.

Method Module		MRR	Hits@1	Hits@10
MKGE	Full Model	0.475	0.387	0.623
	w/o Entity Prior	0.466	0.387	0.613
	w/o Asymmetric Relational Fingerprint	0.449	0.377	0.589
	w/o Multi-Relation Aggregation	0.445	0.353	0.584
	w/ Mean Fusion	0.439	0.353	0.599
MPT	Full Model	0.514	0.426	0.670
	w/o Structural Embedding	0.440	0.333	0.633
	w/o R. Context Enhancement	0.439	0.348	0.599
	w/ Concat Fusion	0.427	0.348	0.575

5 Experimental Setup

Datasets. To better evaluate our model on multimodal analogical reasoning tasks, we evaluate RMAR on two multimodal analogical reasoning datasets, **MCNetAnalogy** and **MARS** [38], each paired with its MMKG (**MCNetKG** and **MarKG** [38]) to provide the required structural context. MCNetAnalogy, introduced in this work, builds on ConceptNet [28] by extending the unimodal analogical reasoning dataset *Conceptnet_Relational_Similarity* [31] by augmenting it with rich image data sourced from Unsplash¹ and Google Images. To remedy MARS’s sparse coverage of abstract concepts, it deliberately includes abstract and action entities alongside concrete ones, linked via diverse relation templates. Detailed statistics for both datasets are summarized in Table 1.

Evaluation Metrics. To assess model performance, we formulate the analogical reasoning task as link prediction following KG completion [3]. Following established protocols, we adopt rank-based metrics: Mean Reciprocal Rank (MRR) and Hits at k (Hits@k). MRR averages the reciprocal rank of the gold tail, while Hits@k measures the fraction of gold tails ranked in the top-k predictions.

¹<https://unsplash.com/>

Baselines. For a fair and comprehensive evaluation, we compare RMAR against nine strong baselines across MKGE and MPT.

- **MKGE Methods.** **TransAE** [35] enhances translational KGE with multimodal features; **IKRL** [36] builds instance-level image representations aligned with entities; **RSME** [25] supplies contrastive vision language encoders for entity semantics.
- **MPT Methods.** **ViLT** [20] is a vision–language Transformer; **ViLBERT** [22] uses dual stream co-attention; **VisualBERT** [21] integrates visual regions into a single stream encoder; **FLAVA** [27] unifies multi-task multimodal pre-training; **MKGFormer** [7] adapts Transformers to MMKG reasoning; **TIMAR** [5] enhances MPTs with task-aware multimodal alignment.

Implementation Details. Our proposed **RMAR** model is implemented in two distinct settings. In the **MKGE setting**, our model is built upon RotatE [29], a KGE model that represents relations as rotations in a complex vector space. In the **MPT setting**, the implementation is based on the MKGFormer architecture [38]. All our experiments are conducted on a Linux server running Ubuntu 24.04.3 LTS, equipped with a single NVIDIA RTX 4090 GPU. Detailed hyperparameter settings are provided in Appendix C.1.

6 Experimental Results

We conduct comprehensive quantitative and qualitative experiments to evaluate RMAR framework.

6.1 Main Results

The main results are reported in Table 2. RMAR delivers substantial gains over state-of-the-art (SOTA) baselines on both datasets. On **MCNetAnalogy**, designed to challenge models with abstract, action, concrete entities and diverse relations, **RMAR-MKGFormer** achieves new SOTA across all metrics, improving MRR by 16.6% over the strongest MPT baseline. Our lightweight MKGE variant, **RMAR-RotatE**, is also highly effective, improving MRR by 31.6% over the second-best MKGE method and outperforming strong

Reasoning Setting	Analogy Context	Question Context	Top-3 Tail Entity				Gold Rank
Single Analogical Reasoning (T_h, T_t): ($V_q, ?$)			RMAR	lung cancer	have sex	eat	1
			MKGFormer	get money	gain energy	graduation	50
Blended Analogical Reasoning (V_h, T_t): ($V_q, ?$)			RMAR	marry	find a mate	put on cloth	1
			MKGFormer	put on clothe	a cinema	find a mate	215

Figure 3: Case studies of multimodal analogical reasoning on MCNetAnalogy dataset. Each case study details the analogical context and query context, including their inferred relations.

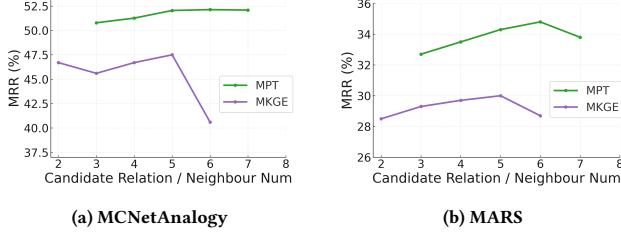


Figure 4: Ablation study of the number of Candidate Relation / Neighbor in MRR for (a) MCNetAnalogy and (b) MARS.

MPT baselines such as FLAVA by 7.7% in MRR, highlighting strong structural modeling without extensive pre-training.

On **MARS**, RMAR remains competitive: **RMAR-MKGFormer** attains the best MRR and Hits@1. Although TIMAR performs better on Hits@k for larger k, we attribute this to RMAR producing a sharper score distribution optimized for Hits@1. Overall, the consistent gains over the respective backbones (RotatE and MKGFormer) on both datasets validate the effectiveness of RMAR.

6.2 Ablation Study

To validate each key component of RMAR, we conduct an ablation study (Table 3). The results confirm that every module contributes to the final performance; additional MARS ablations are reported in Appendix D.1. For the **MKG-based model**, removing entity- or relation-level modality preference reduces performance, with a larger drop when ablating relation preference, supporting the effectiveness of relation-aware fusion. The largest degradation occurs without the Candidate Relation module, indicating that modeling multiple candidate relations is essential for analogical reasoning. For the **MPT-based model**, removing structural information causes a clear decline, and removing Relational Enhancement leads to an even larger drop, highlighting its critical role. We also compare our dynamic fusion to common alternatives: replacing Modality Preference with mean-pooling or Gated fusion with concatenation consistently hurts performance, demonstrating the advantage of our dynamic mechanisms for multimodal integration.

6.3 Sensitivity Study

We analyze the effect of the number of candidate relations for MKGE and candidate neighbors for MPT by varying K . Figure 4 reports performance on MCNetAnalogy and MARS as K ranges from 2 to 7. Across both datasets, performance increases with K

up to an optimum and then declines. MKGE consistently peaks at $K = 5$, while MPT performs best at $K = 6$. This suggests a trade-off: small K yields an insufficiently diverse candidate set, whereas large K introduces many less relevant candidates, adding noise and degrading reasoning.

6.4 Case Study

To qualitatively demonstrate the advantages of our approach, Figure 3 presents two case studies comparing RMAR with MKGFormer on abstract and causal reasoning. The first case, focuses on causal reasoning, inferring *lung cancer* from *smoking*. Visual evidence of smoking does not convey this long-term medical outcome, leaving MKGFormer unable to bridge the action–consequence gap. RMAR succeeds by leveraging explicit causal pathways in the KG that connect the two concepts. In the second case, the target concept *marry* is an abstract social contract: wedding images capture only the ceremony and miss the deeper social and legal meanings. As a result, MKGFormer, which relies heavily on semantic features, struggles to link the query to the correct abstract concept. RMAR instead exploits KG structure, recognizing *marrying* as a prerequisite for *starting a family*, and thus makes the correct inference. Overall, these examples show that reasoning beyond semantic similarity requires deep structural knowledge, supporting RMAR’s adaptive fusion. More case studies are provided in Appendix D.3.

7 Conclusion

We present RMAR, a relation-aware framework that learns modality-compatible representations from analogical context via explicit relational fingerprints and implicit gated fusion. RMAR couples abductive relation induction with inductive tail completion, preserving the analogical signal while aggregating evidence. We also release MCNETANALOGY and MCNETKG to broaden evaluation toward abstract concepts and diverse relations. Across MCNETANALOGY and MARS, RMAR establishes new state-of-the-art results, including up to 29% relative improvement on MCNETANALOGY, validating that relation-aware preference learning is more principled and effective than static fusion. Future work includes zero-shot transfer across knowledge graphs and cross-domain generalization.

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Appendices

A Methodological Details

A.1 Explicit Fingerprint-based Algorithm

Algorithm 1 summarizes the explicit fingerprint-based approach.

Algorithm 1 RMAR: Explicit Fingerprint-based Approach

```

1: Input: Analogical query  $((e_h, e_t), e_q)$ , entity set  $\mathcal{E}$ , relation set  $\mathcal{R}$ , pre-computed entity priors  $\{\mathbf{w}_e\}$ , relational fingerprints  $\{(\mathbf{w}_{rh}, \mathbf{w}_{rt})\}$ , KGE scorer  $s_{\text{structure}}$ .
2: Output: Predicted tail entity  $\hat{e}_t$ .

3: — Phase 1: Abductive Relation Induction —
4: for each relation  $r \in \mathcal{R}$  do
5:   ▶ Calculate modality compatibility score based on entity priors and relation fingerprints
6:    $s_{\text{modal}}(r) \leftarrow p(\mathbf{w}_{e_h} | \mathbf{w}_{rh}) \times p(\mathbf{w}_{e_t} | \mathbf{w}_{rt})$ 
7:   ▶ Combine with structural plausibility score
8:    $s_{\text{final}}(r) \leftarrow s_{\text{structure}}(e_h, r, e_t) + \log(s_{\text{modal}}(r))$ 
9: end for
10:  $\mathcal{R}_K \leftarrow \text{Top-}K_{r \in \mathcal{R}}(s_{\text{final}}(r))$  ▶ Select Top-K candidate relations
11:  $\{\omega_{r'}\}_{r' \in \mathcal{R}_K} \leftarrow \text{Softmax}(\{s_{\text{final}}(r')\}_{r' \in \mathcal{R}_K})$  ▶ Calculate relation confidence weights

12: — Phase 2: Inductive Tail Completion —
13: for each candidate tail entity  $e' \in \mathcal{E}$  do
14:    $s_{q_t}(e') \leftarrow 0$ 
15:   for each relation  $r' \in \mathcal{R}_K$  do
16:     ▶ Entity-centric view: fuse embeddings using entity priors
17:      $\mathbf{h}_{e_q} \leftarrow \sum_m w_{e_q}^m \mathbf{h}_{e_q}^{(m)}; \quad \mathbf{h}_{e'} \leftarrow \sum_m w_{e'}^m \mathbf{h}_{e'}^{(m)}$ 
18:      $s_{\text{entity}} \leftarrow s_{\text{structure}}(\mathbf{h}_{e_q}, \mathbf{h}_{r'}, \mathbf{h}_{e'})$ 
19:     ▶ Relation-centric view: fuse embeddings using relation fingerprints
20:      $\mathbf{h}'_{e_q} \leftarrow \sum_m w_{r'h}^m \mathbf{h}_{e_q}^{(m)}; \quad \mathbf{h}'_{e'} \leftarrow \sum_m w_{r't}^m \mathbf{h}_{e'}^{(m)}$ 
21:      $s_{\text{relation}} \leftarrow s_{\text{structure}}(\mathbf{h}'_{e_q}, \mathbf{h}_{r'}, \mathbf{h}'_{e'})$ 
22:     ▶ Aggregate scores from both views, weighted by relation confidence
23:      $s_{\text{dual}} \leftarrow \lambda \cdot s_{\text{entity}} + (1 - \lambda) \cdot s_{\text{relation}}$ 
24:      $s_{q_t}(e') \leftarrow s_{q_t}(e') + \omega_{r'} \cdot s_{\text{dual}}$ 
25:   end for
26: end for
27:  $\hat{e}_t \leftarrow \arg \max_{e' \in \mathcal{E}} s_{q_t}(e')$ 
28: return  $\hat{e}_t$ 

```

A.2 Implicit Gating-based Algorithm

Algorithm 2 summarizes the implicit gating-based approach.

A.3 Relation Memory for Candidate Sampling

While in-batch neighbor sampling provides dynamic, context-specific examples, it is inherently limited by the batch size. A small batch may suffer from sparse neighbors or a lack of high-quality semantically similar instances. To address this, we incorporate two key components: a **Memory Bank** and a **Momentum Encoder**.

Algorithm 2 RMAR: Implicit Gating-based Approach

```

1: Input: Analogical query  $((e_h, e_t), e_q)$ , trained MPT model  $f_q$ , memory bank  $\mathcal{M}$ .
2: Output: Predicted tail entity  $\hat{e}_t$ .

3: — Phase 1: Abductive Context Enhancement —
4: ▶ Process the example pair to generate the enhanced relation vector
5:  $(\mathbf{h}_{h,\text{ex}}, \mathbf{h}_{t,\text{ex}}, \mathbf{h}_{[R],\text{ex}}) \leftarrow f_q((e_h, e_t))$ 
6:  $\mathbf{r}_{\text{emb}} \leftarrow \mathbf{h}_{h,\text{ex}} - \mathbf{h}_{t,\text{ex}}$ 
7:  $\mathcal{N} \leftarrow \text{FindNeighbors}(\mathbf{r}_{\text{emb}}, \mathcal{M})$ 
8:  $\mathbf{h}_{[R],\text{coarse}} \leftarrow \text{Attention}(\mathbf{h}_{[R],\text{ex}}, \{\mathbf{h}_{[R],k}\}_{k \in \mathcal{N}})$ 
9:  $\mathbf{h}_{[R],\text{enhanced}} \leftarrow \mathbf{h}_{[R],\text{ex}} + \mathbf{h}_{[R],\text{coarse}}$ 

10: — Phase 2: Inductive Tail Prediction —
11: ▶ Process the query pair to get logits for the masked position
12:  $\text{logits}_{\text{mask}} \leftarrow f_q((e_q, [\text{MASK}]))$ 

13: ▶ The encoder  $f_q$ , trained to align with  $\mathbf{h}_{[R],\text{enhanced}}$ , implicitly uses this relational understanding for prediction.
14:  $\hat{e}_t \leftarrow \arg \max_{e' \in \mathcal{E}} (\text{logits}_{\text{mask}}[e'])$  ▶ Select the top-scoring entity
15: return  $\hat{e}_t$ 

```

A.3.1 Memory Bank. To expand the scope of the neighborhood search, we introduce a memory bank, $\mathcal{M}$, implemented as a large First-In, First-Out (FIFO) queue with a capacity of K , which is significantly larger than the batch size. This memory bank stores pairs of relational embeddings and their corresponding "[R]" token representations from recent training batches. When constructing the neighborhood for an anchor, we sample from both the current batch and this memory bank, providing a much richer and more diverse set of candidate relations.

A.3.2 Momentum Encoder. A challenge in using a memory bank is ensuring the consistency of the features. The main model encoder is updated rapidly via backpropagation, so a representation from an early batch may not be comparable to one from a later batch. To solve this, we create a momentum-updated "twin" of our main model, which we call the **momentum encoder**.

Let the parameters of the main online encoder (the one updated by backpropagation) be denoted as θ_q , and the parameters of the momentum encoder be θ_k . While the optimizer updates θ_q , the parameters of the momentum encoder θ_k are a slowly evolving exponential moving average (EMA) of θ_q . After each training step, θ_k is updated as follows:

$$\theta_k \leftarrow m \cdot \theta_k + (1 - m) \cdot \theta_q \quad (23)$$

where $m \in [0, 1)$ is a momentum coefficient. The momentum encoder is not updated via backpropagation.

All representations stored in the memory bank are computed using this stable momentum encoder. During the neighborhood search, the anchor's relational embedding $\mathbf{r}_{\text{emb}}^{(i)}$ is computed with the online encoder, while the candidate keys in the memory bank have been computed with the momentum encoder. This ensures that the keys in the memory bank are consistent with each other

over time, creating a stable and reliable dictionary for the similarity search. After the attention and loss computation, the current batch’s representations from the momentum encoder are enqueued to the memory bank. The Ablation Study for Relation Memory is in Appendix D.2.

A.4 Robustness to Representation Perturbations

To explicitly probe the representation-quality issue, we evaluate RMAR under degraded modality features by injecting synthetic noise into the image and text embeddings before multimodal fusion, while keeping the KG structural signals unchanged. Beyond the standard RMAR, we further introduce a lightweight anti-noise plug-in as a robustness variant: we Gaussianize modality embeddings and compute fusion weights from per-modality uncertainty (variance) rather than raw preference scores. This uncertainty-aware fingerprint + gating mechanism down-weights modalities with higher estimated uncertainty, allowing the model to rely more on higher-quality evidence and the relational structure encoded in the KG. Even in this more challenging setting, RMAR-RotatE on **MARS** remains robust and achieves **28.52%** MRR, indicating that relational fingerprints preserve useful signal despite perturbed modality representations. We will release these robustness experiments and implementation details with the code.

B Additional Datasets Information

B.1 Dataset Construction

To address the limitations of existing benchmarks, particularly their scarcity of abstract concepts, we constructed a new multimodal knowledge graph, **MCNetKG**, and its corresponding analogical reasoning dataset, **MCNetAnalogy**. The construction process followed three main stages: seed analogy collection, knowledge graph expansion, and multimodal enrichment.

B.1.1 Step 1. Seed Analogy Data Collection. The foundation of our benchmark is a high-quality set of textual analogies. We began with the *ConceptNet Relational Similarity* [31] dataset, a question-answer dataset containing entities and analogical relations extracted from ConceptNet [28]. To ensure the integrity of our subsequent graph expansion, we first performed a filtering step, retaining only those entities that could be successfully linked back to the ConceptNet knowledge base. This initial process yielded a high-quality **Seed Analogy Dataset** comprising 1,663 entities, 18 relations, and 2,304 analogical triples.

B.1.2 Step 2. Knowledge Graph Expansion. Reasoning with semantic information alone neglects the rich structural context inherent in a knowledge graph. To incorporate this, we utilized the Seed Analogy Dataset as a reference point to expand the underlying graph structure. We linked the seed entities to the ConceptNet [28] database and performed a controlled two-hop neighborhood expansion. To maintain quality and relevance, we limited the number of 1-hop and 2-hop neighbors to 20 and 5, respectively, and applied a weight threshold to prune low-quality connections. After deduplication and cleaning, this graph densification process resulted in a graph of 9,789 entities, 36 relations, and 18,785 triples.

B.1.3 Step 3. Multimodal Enrichment. The entities derived from ConceptNet typically only contain short phrasal text. To create a truly multimodal resource, we enriched them with long-form textual descriptions and visual imagery.

- **Textual Descriptions:** We implemented a two-stage pipeline for acquiring descriptive text. First, we queried WordNet using the entity’s text and part-of-speech tag to find its formal definition. For entities not found in WordNet, we linked them to Wikipedia and extracted the introductory paragraph as their description. Any entity lacking a description after this process was removed.
- **Visual Imagery:** We collected images using the Unsplash and Google Search APIs. We are grateful to the Unsplash team and their community of photographers for their support. For each entity, we initially fetched up to fifteen images and then applied a rigorous four-stage filtering process to ensure quality and relevance: (1) removal of images with invalid formats; (2) filtering of low-resolution (e.g., smaller than 50x50 pixels) or duplicate images; (3) programmatic removal of visual outliers by using CLIP to ensure semantic consistency between images and their corresponding text; and (4) a final round of manual screening to discard any remaining inappropriate images.

B.1.4 Final Dataset. The above steps resulted in **MCNetKG**, a comprehensive multimodal knowledge graph designed for analogical reasoning research. It contains 1,663 core analogy entities augmented with 7,974 neighbor entities, connected by 18 analogy relations and 16 contextual relations, forming 16,572 triples. A detailed statistical comparison between MCNetKG and MarkG is provided in Table 5.

B.2 Dataset Details

Table 5 presents a statistical overview of our proposed multimodal knowledge graph, **MCNetKG**, alongside three other widely used benchmarks. While datasets such as MarkG and FB15k-IMG are derived from encyclopedic knowledge bases (Wikidata and Freebase), they primarily focus on named, concrete entities like people, places, and organizations. WN9-IMG, based on the lexical database WordNet, centers on lexicosemantic relations. In contrast, **MCNetKG** is unique in its foundation on ConceptNet, a commonsense knowledge graph. This allows our benchmark to feature a much richer set of **abstract concepts, actions, and commonsense relational patterns** (e.g., *HasProperty*, *CapableOf*), which are underrepresented in other datasets. This focus on abstract knowledge provides a more challenging and comprehensive testbed for evaluating a model’s capacity for human-like analogical reasoning.

C Experimental Details

C.1 Hyperparameter Settings

For the MKGE family, we train the model for 1000 epochs with a fixed batch size of 1024. The model is optimized using Adagrad. We perform a grid search for the optimal hyperparameters: the embedding dimension is selected from {128, 256}, the number of negative samples K from {16, 32}, the scoring margin γ from {8, 12}, and the learning rate from $\{1e^{-3}, 2e^{-3}\}$. The balance hyperparameter is set to 0.5. For the MPT setting, the model is trained for 30 epochs with a maximum sequence length of 128. We search for the batch size in

Table 4: Case Study on MCNetAnalogy, comparing the final rank of the correct entity for RMAR and the MKGFormer baseline.

Analogy Context	Relation	Question Context	Correct Answer	RMAR Rank	MKGFormer Rank
(happiness, sing)	Causes	(loving, ?)	forgive someone	1	2
(A death, life)	HasFirstSubevent	(God, ?)	religion	1	3
(have sex, good)	Entails	(lemon, ?)	sour	1	184
(night, dark)	Entails	(water, ?)	essential to all life	1	96
(harmonica, make music)	HasPrerequisite	(fire extinguisher, ?)	put out a fire	1	71

Table 5: Statistical comparison of multimodal knowledge graph datasets. The proposed MCNetKG is highlighted.

Dataset	# Entity	# Relation	# Triple	# Image	Source
MCNetKG	9,637	34	16,572	41,417	ConceptNet
MarKG	11,292	192	34,420	76,424	Wikidata
WN9-IMG	6,555	9	14,319	65,550	WordNet
FB15k-IMG	11,757	1,231	350,293	107,570	Freebase

{16, 32} and the learning rate in $\{3e^{-5}, 5e^{-5}\}$. The balance hyperparameter is set to 0.43. For all baseline models, we adopt the optimal hyperparameters reported in their original publications to ensure a fair comparison. We report results for some baselines directly from prior work, when the setups are identical to ours [5, 38].

D Additional Experiments

D.1 Additional Ablation Study on MARS

Table 6: Ablation study of RMAR on the MARS dataset.

Method	Module	MRR	Hits@1	Hits@10
MKGE	Full Model	0.300	0.268	0.356
	w/o Entity Prior	0.272	0.251	0.325
	w/o Asymmetric Relational Fingerprint	0.269	0.258	0.303
	w/o Multi-Relation Aggregation	0.261	0.244	0.301
	w/ Mean Fusion	0.259	0.236	0.312
MPT	Full Model	0.348	0.319	0.393
	w/o Structure	0.336	0.311	0.385
	w/o R Enhance	0.330	0.305	0.381
	w/ Concat Fusion	0.321	0.294	0.364

The ablation study results on the MARS dataset, presented in Table 6, further validate the contributions of our proposed modules. For the **MKGE-based model**, we observe that replacing our relation-aware fusion with a simple *Mean Fusion* results in the most substantial performance degradation, confirming the effectiveness of our dual-view prediction strategy. The removal of other components, such as *Multi-Relation Aggregation*, also significantly harms performance, underscoring their importance. For the **MPT-based model**, replacing our *Adaptive Gated Fusion* with a standard *Concat Fusion* causes the most severe drop in performance across all metrics. This strongly validates the superiority of our dynamic gating mechanism over simpler fusion alternatives. The removal of structural information and the *R Enhance* module also negatively impacts the results, confirming their critical roles. These trends are consistent with our primary findings on MCNetAnalogy and

demonstrate the general effectiveness of our RMAR framework’s components.

D.2 Ablation Study for Relation Memory

Table 7: Ablation study on the effect of the Relation Memory module in our MPT-based model.

Dataset	Module	MRR	Hits@1	Hits@10
MCNetAnalogy	Full Model	0.514	0.426	0.670
	w/o Relation Memory	0.473	0.382	0.657
MARS	Full Model	0.348	0.319	0.393
	w/o Relation Memory	0.339	0.308	0.394

To prove the contribution of our proposed Relation Memory mechanism, we conduct an ablation study where we disable both the Memory Bank and the Momentum Encoder, forcing the model to find neighbors only within the current batch. The results, presented in Table 7, show a clear degradation in performance on both datasets. The drop is particularly pronounced on our more challenging MCNetAnalogy benchmark, with a substantial decrease in both MRR and Hits@1. This suggests that the larger and more diverse pool of candidate neighbors provided by the memory bank is critical for learning a robust, coarse-grained relation representation. The performance drop on MARS further confirms that relying solely on transient in-batch neighbors is insufficient. This validates our design choice to incorporate a large, consistent candidate pool via the Relation Memory to enhance the context aggregation process.

D.3 More Case Study

To provide a more fine-grained understanding of our model’s capabilities, we present a diverse set of five case studies in Table 4. These examples demonstrate RMAR’s consistent success in reasoning about abstract, causal, and functional relationships where the baseline model struggles. For instance, RMAR correctly identifies nuanced social actions (e.g., *loving* $\rightarrow$ *forgive someone*), highly abstract concepts (*God* $\rightarrow$ *religion*), and functional purposes (*fire extinguisher* $\rightarrow$ *put out a fire*). In these cases, where MKGFormer is either misled by superficial associations leading to a high-ranking error, or provides a plausible but less precise alternative, RMAR’s superiority is evident. This confirms that RMAR’s relation-aware fusion mechanism successfully learns to prioritize the most informative modalities for a given context, leading to more robust and human-like analogical reasoning.